



**Barker**  
College

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Student Number

2021

TRIAL HIGHER SCHOOL  
CERTIFICATE EXAMINATION

# Mathematics Extension 2

## Section I - Multiple Choice

Write your student number at the top of this page.

Select the alternative A, B C or D that best answers the question. Fill in the response oval completely.

Sample  $2 + 4 =$  A 2 B 6 C 8 D 9  
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐  
*correct*

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Start  
Here →

- |                                                                                                    |                                                                                                     |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> | 6. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/>  |
| 2. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> | 7. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/>  |
| 3. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> | 8. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/>  |
| 4. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> | 9. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/>  |
| 5. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> | 10. A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> |

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Student Number

2021

TRIAL HIGHER SCHOOL  
CERTIFICATE EXAMINATION

# Mathematics Extension 2

Staff Involved:

BHC ARM\*

40 copies

8:30 AM  
FRIDAY 20 AUGUST

## TOPICS COVERED

- Complex Numbers I & II
- Proof
- Integration
- Vectors
- Mechanics

## General

### Instructions:

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated

Total marks:  
100

### Section I – 10 marks (pages 4-8)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

### Section II – 90 marks (pages 9-14)

- Attempt Questions 11-16
- Allow about 2 hour and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 What is the negation of the following statement?

If today is Friday, then I have a maths exam.

- A. If I have a maths exam, then today is Friday.
- B. Today is not Friday and I do not have a maths exam.
- C. If I do not have a maths exam, then today is not Friday.
- D. Today is Friday and I do not have a maths exam.

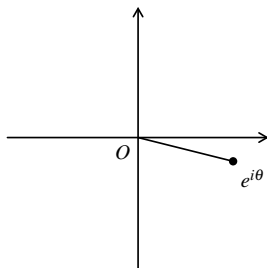
- 2 Given the vectors  $\underline{a} = 3\underline{i} - 4\underline{j} + 12\underline{k}$  and  $\underline{b} = 2\underline{i} + 2\underline{j} - \underline{k}$ , what is the projection of  $\underline{a}$  onto  $\underline{b}$ ?

- A.  $-\frac{14}{3}(2\underline{i} + 2\underline{j} - \underline{k})$
- B.  $-\frac{14}{9}(2\underline{i} + 2\underline{j} - \underline{k})$
- C.  $-\frac{14}{13}(3\underline{i} - 4\underline{j} + 12\underline{k})$
- D.  $-\frac{14}{169}(3\underline{i} - 4\underline{j} + 12\underline{k})$

- 3 A particle has a displacement of  $3\underline{i} + \underline{j}$  metres at a particular time. The particle moves with constant velocity and two seconds later its displacement is  $-\underline{i} + 5\underline{j}$  metres. What is the velocity of the particle in  $\text{m.s}^{-1}$ ?

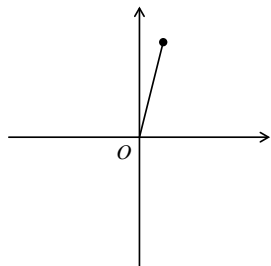
- A.  $2\underline{i} + 6\underline{j}$
- B.  $-4\underline{i} + 4\underline{j}$
- C.  $4\underline{i} - 4\underline{j}$
- D.  $-2\underline{i} + 2\underline{j}$

- 4 The Argand diagram shows the complex number  $e^{i\theta}$ .

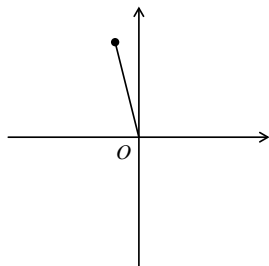


Which of the following diagrams best shows the complex number  $ie^{-i\theta}$ ?

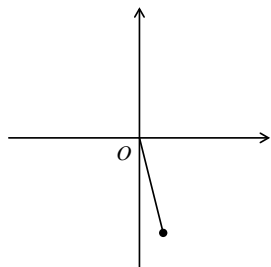
A.



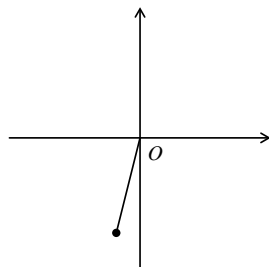
B.



C.



D.



- 5 Given that  $a$  and  $b$  are integers, which one of the following statements has a converse that is always true?

- A. If  $a$  and  $b$  are even, then  $a + b$  is even
- B. If  $a$  is divisible by 8, then  $a$  is even
- C. If  $a$  and  $b$  are odd, then  $a \times b$  is odd
- D. If  $a$  and  $b$  are odd, then  $a - b$  is even

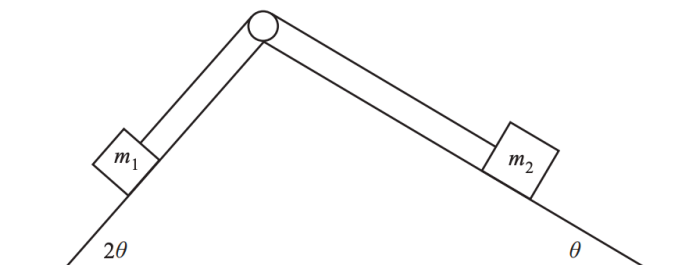
- 6 Which complex number is a cube root of  $-i$ ?

- A.  $\frac{\sqrt{3}}{2} - \frac{1}{2}i$
- B.  $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$
- C.  $\frac{1}{2} - \frac{\sqrt{3}}{2}i$
- D.  $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$

- 7 How many complex numbers satisfy  $|z - 2 - i| = 3$  and  $|z - 6 + 2i| = 2$  simultaneously?

- A. 0
- B. 1
- C. 2
- D. Infinitely many

- 8 Two particles with mass  $m_1$  kg and  $m_2$  kg are connected by a taut light string that passes over a smooth pulley. The particles sit on smooth inclined planes as shown in the diagram below.



If the system is in equilibrium, what is the value of  $\frac{m_1}{m_2}$ ?

- A.  $\frac{\sec \theta}{2}$
- B.  $2 \sec \theta$
- C.  $2 \cos \theta$
- D.  $\frac{1}{2}$
- 9 The integral  $\int \frac{dx}{2 \sin x + 3 \cos x}$  is equivalent to which one of the following expressions?

- A.  $-2 \int \frac{dt}{3t^2 - 4t - 3}$
- B.  $-\int \frac{dt}{3t^2 - 4t - 3}$
- C.  $-2 \int \frac{dt}{3t^2 - 2t - 3}$
- D.  $-\int \frac{dt}{3t^2 - 2t - 3}$

- 10 Given that  $\int_a^b f(x) dx > \int_a^b g(x) dx > 0$ , which one of the following must be true?

- A.  $\forall x \in [a, b] (f(x) > g(x))$
- B.  $\forall x \in [a, b] (g(x) > 0)$
- C.  $\exists x \in [a, b] (f(x) > g(x))$
- D.  $\exists x \in [a, b] (g(x) > f(x))$

## Section II

90 marks

Attempt Questions 11–16

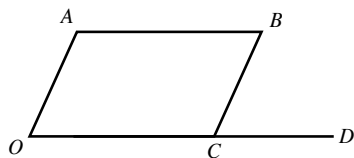
Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

**Question 11** (15 marks) Use a SEPARATE Writing Booklet

- (a) Consider the complex numbers  $w = 1 - 3i$  and  $z = -2 + 5i$ .
- (i) Evaluate  $z\bar{z}$ . 1
- (ii) Express  $\frac{w}{z}$  in the form  $x + iy$  where  $x$  and  $y$  are real numbers. 1
- (b) Evaluate  $\int_1^e \frac{1}{x} \cos(\ln x) dx$ . 2
- Give your answer in simplest exact form.
- (c) Find a vector perpendicular to both  $3\hat{i} - 2\hat{j} + \hat{k}$  and  $4\hat{i} + 2\hat{j} - 5\hat{k}$ . 2
- (d) (i) Find the square root of  $5 - 12i$ . 2
- (ii) Hence, solve  $z^2 + 3z + 1 + 3i = 0$ . 2
- (e) Find  $\int \frac{dx}{\sqrt{3 - 2x - x^2}}$ . 2
- (f)  $OABC$  is a parallelogram. Let  $\vec{OA} = \vec{a}$  and  $\vec{OC} = \vec{c}$ . 3
- $X$  is the midpoint of  $AC$ .
- $D$  lies on  $OC$  produced such that  $\frac{OC}{CD} = k$ .
- If  $\vec{XD} = 3\vec{c} - \frac{1}{2}\vec{a}$ , find the value of  $k$ .



**Question 12** (15 marks) Use a SEPARATE Writing Booklet

- (a) Find the equation, in vector form, of the tangent to the circle  $|y| = 5$  at the point  $(-3, 4)$ . 2
- (b) Using the substitution  $x = 4\sin^2 \theta$ , evaluate  $\int_0^2 \sqrt{\frac{x}{4-x}} dx$ . 3
- Give your answer in simplest exact form.
- (c) For positive real numbers  $a$  and  $b$ , prove that  $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$ . 2
- (d) Consider the following statement,  $P$ , for integers  $x$  and  $y$ :
- $P$ : If  $x^2 + y^2$  is divisible by 4 then  $x$  and  $y$  are not both odd.
- (i) State the contrapositive of  $P$ . 1
- (ii) Hence, prove  $P$  by using the contrapositive. 2
- (iii) Without providing a proof, state the conditions on  $x$  and  $y$  for  $x^2 + y^2$  to be divisible by 4. 1
- (e) Use a suitable substitution to evaluate  $\int_1^9 \frac{dx}{x(2\sqrt{x}-1)}$ . 4
- Give your answer in simplest exact form.

**Question 13** (14 marks) Use a SEPARATE Writing Booklet

- (a) On an Argand diagram, the point  $P$  representing the complex number  $z$  moves such that  $|z - (1 + i)| = 1$ .
- (i) Sketch the locus of  $P$ . 1
- (ii) Find the greatest value of  $|z|$ . 1
- (iii) Shade the region where  $|z - (1 + i)| \leq 1$  and  $0 \leq \text{Arg}(z - 1) \leq \frac{\pi}{4}$ . 2
- (iv) Find the area of the region in (iii). 1
- (b) A ball of mass  $m$  kg is thrown vertically downwards off a very high cliff with speed  $20 \text{ m.s}^{-1}$ . It experiences acceleration due to gravity of  $10 \text{ m.s}^{-2}$  and a force due to air resistance of  $\frac{1}{100}mv^2$  where  $v$  is the velocity of the ball.
- (i) Find the terminal velocity that the ball would reach in  $\text{m.s}^{-1}$ . 1
- (ii) Find the distance through which the ball falls to reach a speed of  $30 \text{ m.s}^{-1}$ . 3  
Give your answer in metres in simplest exact form.
- (c) A particle moves in a straight line. Its displacement,  $x$  metres, after  $t$  seconds is given by  $x = 2 \cos\left(2t - \frac{2\pi}{3}\right) + 3$ .
- (i) Prove the particle is moving in simple harmonic motion about  $x = 3$  by showing that  $\ddot{x} = -4(x - 3)$ . 2
- (ii) What is the period of the motion? 1
- (iii) Find all times within the first  $\pi$  seconds when the particle is moving at 2 metres per second in either direction. 2

**Question 14** (15 marks) Use a SEPARATE Writing Booklet

- (a) A particle of mass  $m$  kg is projected vertically upwards with a velocity of  $\sqrt{\frac{g}{k}} \text{ m.s}^{-1}$  4  
where  $g$  is the magnitude of the acceleration due to gravity. It experiences air resistance of magnitude  $mkv^2$ . Show that the time taken to reach its maximum height is  $\frac{\pi}{4\sqrt{gk}}$ .
- (b) The point  $A$  has a position vector  $\begin{pmatrix} -2 \\ 4 \\ 7 \end{pmatrix}$  and the point  $B$  has position vector  $\begin{pmatrix} -1 \\ 3 \\ 8 \end{pmatrix}$ .
- (i) Find a vector equation for  $\ell_1$ , the line passing through  $A$  and  $B$ . 2
- (ii) The point  $P$  has position vector  $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ . 2  
If  $\angle PBA = \theta$ , show that  $\cos \theta = \frac{1}{3}$ .
- (iii) The line  $\ell_2$  passes through the point  $P$  and is parallel to the line  $\ell_1$ . 1  
Find a vector equation for  $\ell_2$ .
- (iv) The points  $C$  and  $D$  both lie on  $\ell_2$ . Given that  $AB = PC = DP$  and that the  $x$ -coordinate of  $C$  is positive, find the coordinates of  $C$  and  $D$ . 3
- (v) Find the exact area of trapezium  $ABCD$ , giving your answer as a simplified surd. 3

**Question 15** (17 marks) Use a SEPARATE Writing Booklet

- (a) A particle is moving such that  $\ddot{x} = 18x^3 + 27x^2 + 9x$ .  
Initially,  $x = -2$  and the velocity,  $v = -6$ .
- (i) Show that  $v^2 = 9x^2(1+x)^2$ . 2
- (ii) Hence, or otherwise, show that  $\int \frac{1}{x(1+x)} dx = -3t$ . 2
- (iii) Hence, find  $x$  as a function of  $t$ . 4
- (b) Find  $\int (1+2x^2)e^{x^2} dx$ . 3
- (c) (i) Use de Moivre's Theorem to show that 3
- $$\tan 7\theta = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6} \quad \text{where } t = \tan \theta$$
- (ii) Hence, show that the roots of the equation  $x^3 - 21x^2 + 35x - 7 = 0$  are 3
- $$\tan^2\left(\frac{\pi}{7}\right), \tan^2\left(\frac{2\pi}{7}\right) \text{ and } \tan^2\left(\frac{3\pi}{7}\right)$$

**Question 16** (14 marks) Use a SEPARATE Writing Booklet

- (a) (i) For a function  $g(x)$ , continuous on  $[a, b]$ , prove that 2
- $$\int_a^b g(x) dx = \int_a^b g(a+b-x) dx$$
- (ii) Hence, or otherwise, for an **even** function  $f(x)$ , continuous on  $[-a, a]$ , show that 3
- $$\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$$
- (b) For integers  $n \geq 0$ , let  $I_n(x) = \int_0^x t^n e^{-t} dt$ .
- (i) Prove by induction that 4
- $$I_n(x) = n! \left[ 1 - e^{-x} \left( 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} \right) \right]$$
- (ii) Show that 2
- $$0 \leq \int_0^1 t^n e^{-t} dt \leq \frac{1}{n+1}$$
- (iii) Hence, show that 1
- $$0 \leq 1 - e^{-1} \left( 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} \right) \leq \frac{1}{(n+1)!}$$
- (iv) Hence, find the limiting value of  $1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$  as  $n \rightarrow \infty$ . 2
- Show working to justify your answer.

**End of paper**

# Year 12 Extension 2 Trial Exam

Q1, "If today is Friday, then I have a maths exam" is of the form

$$F \Rightarrow E$$

which is logically equivalent to

$$\sim F \vee E.$$

The negation of  $F \Rightarrow E$  is

$$\sim(\sim F \vee E)$$

which is equivalent to

$$F \wedge \sim E,$$

namely,

"It is Friday and I don't have a maths exam."

(d)

Q2, Now  $\text{proj}_{\vec{b}} \vec{a}$

$$= \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \times \vec{b}$$

$$= \frac{6-8-12}{(\sqrt{9})^2} \times (2\vec{i} + 2\vec{j} - \vec{k})$$

$$= -\frac{14}{9} (2\vec{i} + 2\vec{j} - \vec{k})$$

(b)

Q3, The displacement in 2 sec is

$$(\vec{i} + 5\vec{j}) - (3\vec{i} + \vec{j}) = -4\vec{i} + 4\vec{j}$$

Hence the velocity in m/s is

$$\frac{-4\vec{i} + 4\vec{j}}{2} = -2\vec{i} + 2\vec{j}$$

(d)

Q4, Let  $\alpha = -\frac{\pi}{6}$  such that

$$e^{i\alpha} = e^{i(-\pi/6)} \text{ then}$$

$$ie^{i\alpha} = ie^{i(-\pi/6)}$$

$$= i\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$$= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(b)

Q5, The converse of  $p \Rightarrow q$  is

$$q \Rightarrow p.$$

The converse of (c) is

"If  $a \times b$  is odd, then

$a$  and  $b$  are odd."

This is always true.

$$Q6, \vec{z}^3 = -i$$

$$= \cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)$$

$$\therefore \vec{z} = \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)$$

$$\vec{z} = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

(a)

$$Q7, |z - (2+i)| = 3$$

is a circle centre  $(2,1)$   
radius 3 units.

$$|z - (6-2i)| = 2$$

is a circle centre  $(6,-2)$   
radius 2 units.

Consider the distance between the two centres:

$$(2,1) \quad d = \sqrt{16+9} = 5$$

$$\text{But } 3+2=5$$

$\Rightarrow$  The two circles touch at one point (b)

$$Q8, \int \frac{dx}{2\sin x + 3\cos x} \quad dx$$

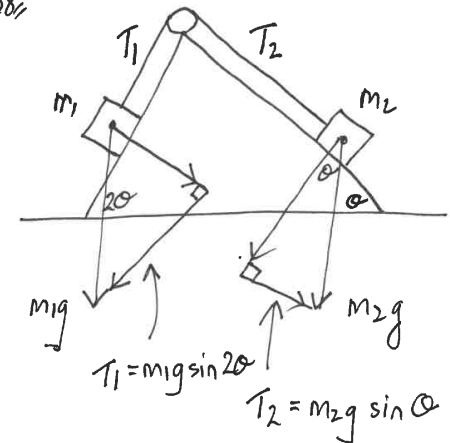
$$\text{Let } t = \tan\left(\frac{x}{2}\right)$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$\frac{dt}{dx} = \frac{1}{2}(1+t^2)$$

$$dx = \frac{2dt}{1+t^2}$$

Q8,



$$T_1 = T_2$$

$$\therefore m_1 g \sin 20 = m_2 g \sin \theta$$

$$\therefore \frac{m_1}{m_2} = \frac{\sin \theta}{\sin 20} = \frac{1}{2 \cos \theta}$$

(a)

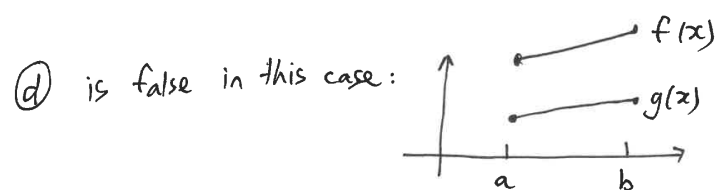
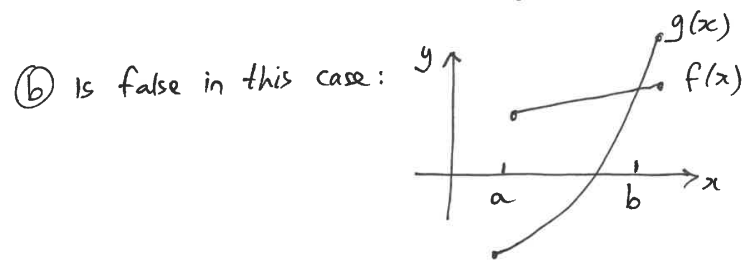
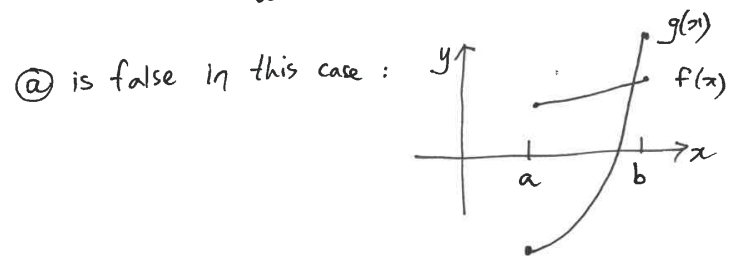
$$I = \int \frac{2dt/1+t^2}{2\left(\frac{2t}{1+t^2}\right) + 3\left(\frac{1-t^2}{1+t^2}\right)}$$

$$= \int \frac{2dt}{4t + 3 - 3t^2}$$

$$= \int \frac{-2dt}{(3t^2 - 4t - 3)}$$

(a)

Q10, Given that  $\int_a^b f(x) dx > \int_a^b g(x) dx > 0$



Therefore Ⓒ  
because if there was not one single  $x$ -value  
for which  $f(x) > g(x)$  then  $f(x)$  lies entirely  
underneath  $g(x)$ , which would make it  
impossible that

$$\int_a^b f(x) dx > \int_a^b g(x) dx > 0.$$

Q11,

(a)  $w = 1-3i$

$$z = -2+5i$$

(i)  $z\bar{z} = (-2+5i)(-2-5i)$   
 $= 29$

(ii)  $\frac{w}{z} = \frac{1-3i}{-2+5i} \times \frac{-2-5i}{-2-5i}$   
 $= \frac{-2-5i+6i-15}{29}$   
 $= \frac{-17}{29} + \frac{1}{29}i$

(b)  $\int_1^e \frac{1}{x} \cos(\ln x) dx$   
 $= \sin(\ln x) \Big|_1^e$   
 $= \sin(1)$   
 $\approx 0.8415$

(c) let the required vector be  
expressed as  $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ . We need

$$3a - 2b + c = 0 \quad \text{and}$$

$$4a + 2b - 5c = 0$$

$$7a - 4c = 0$$

$$a = \frac{4c}{7}$$

If  $c=1$ ,  $a = \frac{4}{7}$  and  $b = \frac{19}{14}$

(d) (i)

let  $(x+iy)^2 = 5-12i$

$$x^2 - y^2 = 5 \quad 2xyi = -12i$$

$$y = \frac{-6}{x}$$

$$x^2 - \left(\frac{-6}{x}\right)^2 = 5$$

$$x^4 - 5x^2 - 36 = 0$$

$$(x^2-9)(x^2+4) = 0$$

$$x = \pm 3$$

when  $x=3$ ,  $y=-2$

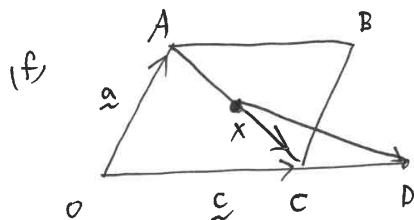
when  $x=-3$ ,  $y=2$

Hence  $\pm(3-2i)$

(ii)  $z^2 + 3z + (1+3i) = 0$   
 $z = \frac{-3 \pm \sqrt{5-12i}}{2}$   
 $z = \frac{-3 \pm (3-2i)}{2}$   
 $z = -i, -3+i$

Hence  $\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 4/7 \\ 19/14 \\ 1 \end{pmatrix}$   
 $= \begin{pmatrix} 8 \\ 19 \\ 14 \end{pmatrix}$

$$\begin{aligned}
 \text{(e)} \quad & \int \frac{dx}{\sqrt{3-2x-x^2}} \\
 &= \int \frac{dx}{\sqrt{3-(x^2+2x)}} \\
 &= \int \frac{dx}{\sqrt{4-(x+1)^2}} \\
 &= \sin^{-1} \left( \frac{x+1}{2} \right) + C
 \end{aligned}$$



$$\begin{aligned}
 \frac{OC}{CD} &= k \Rightarrow CD = \frac{c}{k} \\
 \vec{XD} &= 3\vec{c} - \frac{1}{2}\vec{a} \\
 \text{Now } \vec{AC} &= \vec{c} - \vec{a} \\
 \vec{OX} &= \vec{OA} + \frac{1}{2}\vec{AC} \\
 &= \vec{a} + \frac{1}{2}(\vec{c} - \vec{a}) \\
 &= \frac{1}{2}(\vec{a} + \vec{c}) \\
 \vec{OD} &= \vec{OX} + \vec{XD} \\
 &= \frac{\vec{a}}{2} + \frac{\vec{c}}{2} + 3\vec{c} - \frac{\vec{a}}{2} \\
 &= \frac{7\vec{c}}{2}
 \end{aligned}$$

But  $\vec{OD} = \vec{OC} + \vec{CD}$   
 $= \vec{c} + \frac{\vec{c}}{k}$

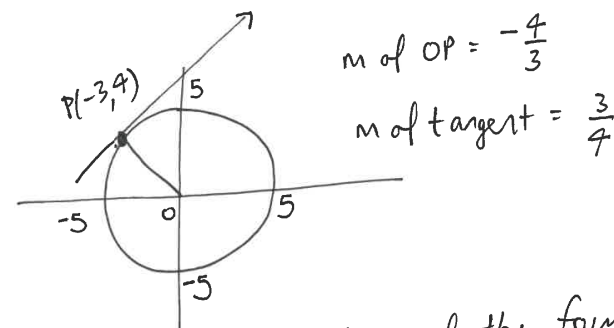
$$\therefore \frac{7\vec{c}}{2} = \vec{c} + \frac{\vec{c}}{k}$$

$$\frac{5\vec{c}}{2} = \frac{\vec{c}}{k}$$

$$\therefore k = \frac{2}{5}$$

Q12

(a)



The tangent has a vector equation of the form  
 $\underline{R} = (-3\hat{i} + 4\hat{j}) + k(4\hat{i} + 3\hat{j})$

$$\text{(b)} \quad \int_0^2 \sqrt{\frac{x}{4-x}} dx$$

$$\begin{aligned}
 \text{Let } x &= 4 \sin^2 \theta \\
 dx &= 8 \sin \theta \cos \theta d\theta
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x=2, \quad \theta &= \frac{\pi}{4} \\
 \text{When } x=0, \quad \theta &= 0
 \end{aligned}$$

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{4}} \sqrt{\frac{4 \sin^2 \theta}{4 - 4 \sin^2 \theta}} \times 8 \sin \theta \cos \theta d\theta \\
 &= 8 \int_0^{\frac{\pi}{4}} \sin^2 \theta d\theta \\
 &= 4 \int_0^{\frac{\pi}{4}} 1 - \cos 2\theta d\theta \\
 &= 4 \left[ \theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}} \\
 &= \pi - 2
 \end{aligned}$$

(c) Prove that  $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$

Consider LHS - RHS

$$= \frac{1}{a} + \frac{1}{b} - \frac{4}{a+b}$$

$$= \frac{b(a+b) + a(a+b) - 4ab}{ab(a+b)}$$

$$= \frac{a^2 - 2ab + b^2}{ab(a+b)}$$

$$= \frac{(a-b)^2}{ab(a+b)}$$

Now the numerator is always  $\geq 0$  and the denominator is always  $> 0$ .

Hence LHS - RHS  $\geq 0$ .

Therefore

$$\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}.$$

(d) P: If  $x^2 + y^2$  is divisible by 4 then  $x$  and  $y$  are not both odd.

(i) The contrapositive of P is:

"If  $x$  and  $y$  are both odd then  $x^2 + y^2$  is not divisible by 4."

(ii) Let  $x = 2k+1$   
 $y = 2j+1$

$$\begin{aligned} \text{then } x^2 + y^2 &= (2k+1)^2 + (2j+1)^2 \\ &= 4k^2 + 4k + 1 + 4j^2 + 4j + 1 \\ &= 4(k^2 + k + j^2 + j) + 2 \end{aligned}$$

which is not divisible by 4.

Hence, if  $x$  and  $y$  are both odd then  $x^2 + y^2$  is not divisible by 4.

Therefore, if  $x^2 + y^2$  is divisible by 4, then  $x$  and  $y$  are not both odd.

(iii) For  $x^2 + y^2$  to be divisible by 4, the condition on  $x$  and  $y$  is that they are both even.

$$(e) \int_1^9 \frac{dx}{x(2\sqrt{x}-1)}$$

$$\text{let } x = u^2 \\ dx = 2u \, du$$

$$\text{when } x=9, \quad u=3 \\ \text{when } x=1, \quad u=1$$

$$I = \int_1^3 \frac{2u \, du}{u^2(2u-1)} \\ = \int_1^3 \frac{2 \, du}{u(2u-1)}$$

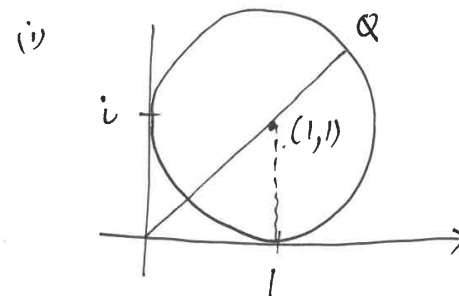
$$\text{let } \frac{2}{u(2u-1)} = \frac{A}{u} + \frac{B}{2u-1}$$

$$2 = A(2u-1) + Bu$$

$$A = -2, \quad B = 4$$

$$I = \int_1^3 \left( \frac{-2}{u} + \frac{4}{2u-1} \right) du \\ = \left[ -2 \ln u + 2 \ln(2u-1) \right]_1^3 \\ = -2 \ln 3 + 2 \ln 5 - 0 \\ = \ln \left( \frac{25}{9} \right)$$

$$Q13, (a) |z - (1+i)| = 1$$

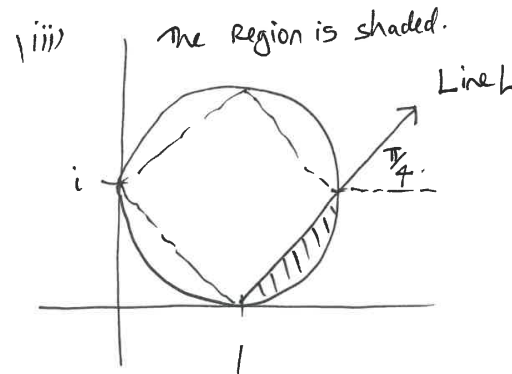


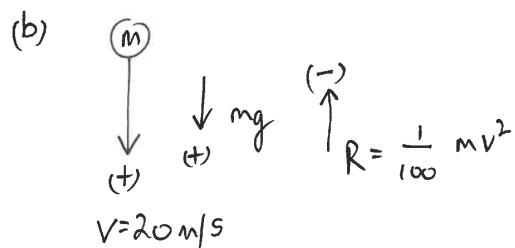
(ii) The greatest value of  $|z|$  occurs when P is at Q.

The greatest value of  $|z|$  is  $1 + \sqrt{2}$

(iv) Line L has equation  $y = x - 1$ . L intersects the circle at (2, 1).

$$\text{Shaded area} \\ = \frac{1}{4} [\pi R^2 - (\sqrt{2})^2] \\ = \frac{1}{4} (\pi - 2)$$





$$(i) \quad m\ddot{x} = 10m - \frac{1}{100} m v^2$$

$$\ddot{x} = 10 - \frac{1}{100} v^2$$

Put  $\ddot{x} = 0$ ,

$$0 = 10 - \frac{1}{100} v^2$$

$$v^2 = 1000$$

$$v = 10\sqrt{10} \text{ m/s.}$$

Terminal velocity is  $10\sqrt{10} \text{ m/s.}$

$$(ii) \quad v \frac{dv}{dx} = 10 - \frac{1}{100} v^2$$

$$\frac{dv}{dx} = \frac{10 - \frac{1}{100} v^2}{v}$$

$$\frac{dv}{dx} = \frac{1000 - v^2}{100v}$$

$$\frac{dx}{dv} = \frac{100v}{1000 - v^2}$$

$$x = -50 \int \frac{-2v}{1000 - v^2} dv$$

$$x = -50 \ln(1000 - v^2) + C$$

But when  $x = 0$ ,  $v = 20$  so

$$C = 50 \ln 600$$

$$\therefore x = 50 \ln \left( \frac{600}{1000 - v^2} \right)$$

When  $v = 30 \text{ m/s,}$

$$x = 50 \ln \left( \frac{600}{100} \right)$$

$$x = 50 \ln(6)$$

$$(c) \quad x = 2 \cos \left( 2t - \frac{2\pi}{3} \right) + 3$$

$$(i) \quad \dot{x} = -4 \sin \left( 2t - \frac{2\pi}{3} \right)$$

$$\ddot{x} = -8 \cos \left( 2t - \frac{2\pi}{3} \right)$$

$$\ddot{x} = -4 \left( 2 \cos \left( 2t - \frac{2\pi}{3} \right) \right)$$

$$\ddot{x} = -4(x - 3)$$

Hence, the particle is moving in SHM about  $x = 3$ .

$$(ii) \quad \text{The period is } \frac{2\pi}{2} = \pi \text{ seconds.}$$

(iii) To find all the times in the first  $\pi$  seconds when the particle is travelling at  $2 \text{ m/s}$  left or right, solve  $2 = -4 \sin \left( 2t - \frac{2\pi}{3} \right)$  and  $-2 = -4 \sin \left( 2t - \frac{2\pi}{3} \right)$

$$-\frac{1}{2} = \sin \left( 2t - \frac{2\pi}{3} \right)$$

$$\frac{1}{2} = \sin \left( 2t - \frac{2\pi}{3} \right)$$

$$\therefore 2t - \frac{2\pi}{3} = \left( \pi + \frac{\pi}{6} \right), \frac{11\pi}{6}, \frac{-\pi}{6} \quad 2t - \frac{2\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$2t = \frac{11\pi}{6}, \frac{3\pi}{6}$$

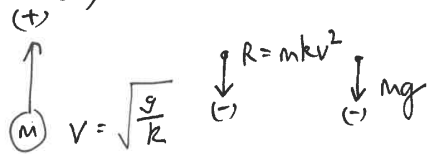
$$2t = \frac{5\pi}{6}, \frac{9\pi}{6}$$

$$t = \frac{11\pi}{12}, \frac{\pi}{4}$$

$$t = \frac{5\pi}{12}, \frac{3\pi}{4}$$

$$\therefore t = \frac{\pi}{4}, \frac{5\pi}{12}, \frac{3\pi}{4}, \frac{11\pi}{12}$$

Q14 (a)



$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -g - kv^2$$

$$\frac{dv}{dt} = -g - kv^2$$

$$\frac{dt}{dv} = \frac{-1}{g + kv^2}$$

$$t = \frac{-1}{k} \int \frac{dv}{(\frac{g}{k}) + v^2}$$

$$t = \frac{-1}{k} \sqrt{\frac{k}{g}} \tan^{-1} \left( \sqrt{\frac{g}{k}} \frac{v}{1} \right) + C$$

But when  $t=0$ ,  $v = \sqrt{g/k}$  so

$$C = \frac{1}{k} \sqrt{\frac{k}{g}} \tan^{-1}(1)$$

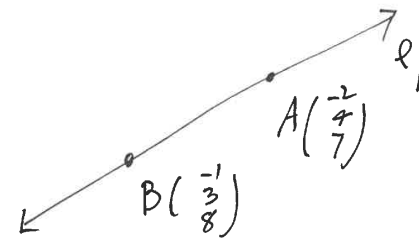
$$C = \frac{1}{\sqrt{gk}} \times \frac{\pi}{4}$$

$$\therefore t = \frac{\pi}{4\sqrt{gk}} - \frac{1}{\sqrt{gk}} \tan^{-1} \left( \sqrt{\frac{g}{k}} \frac{v}{1} \right)$$

when  $v=0$ ,

$$t = \frac{\pi}{4\sqrt{gk}} \text{ as required.}$$

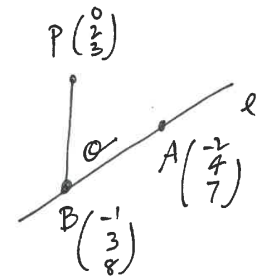
(b) (i)



$$r_1 \text{ is } \vec{OA} + \lambda \vec{AB}$$

$$= \begin{pmatrix} -\frac{2}{7} \\ \frac{1}{7} \end{pmatrix} + \lambda \begin{pmatrix} -\frac{1}{1} \\ \frac{1}{1} \end{pmatrix}$$

(ii)



$$\text{Now } \vec{BP} = \begin{pmatrix} 1 \\ -\frac{1}{5} \end{pmatrix}$$

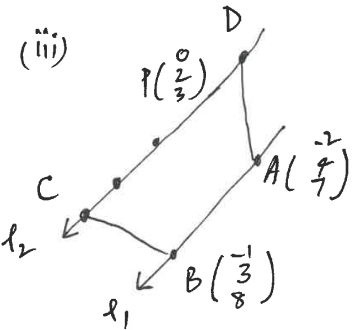
$$\text{and } \vec{BA} = \begin{pmatrix} -\frac{1}{1} \\ \frac{1}{1} \end{pmatrix}$$

$$\therefore \vec{BP} \cdot \vec{BA} = -1 - 1 + 5 = 3$$

$$\therefore 3 = \sqrt{27} \times \sqrt{3} \times \cos \theta$$

$$\therefore \cos \theta = \frac{3}{3\sqrt{3} \times \sqrt{3}}$$

$$\cos \theta = \frac{1}{3} \text{ as required.}$$



$$\text{A vector equation for } l_2 \text{ is}$$

$$= \begin{pmatrix} 0 \\ \frac{2}{3} \end{pmatrix} + \lambda \begin{pmatrix} -\frac{1}{1} \\ \frac{1}{1} \end{pmatrix}$$

(iv) Now  $AB = \sqrt{3}$  so

$$C = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \text{ and}$$

$$D = \begin{pmatrix} -\frac{1}{3} \\ \frac{2}{2} \end{pmatrix}$$

$$(v) |CD| = \sqrt{12}$$

$$|AB| = \sqrt{3}$$

$$h = \sqrt{24}$$

$$\text{Area} = 9\sqrt{2}$$

Q15 (a)

$$\dot{x} = 18x^3 + 27x^2 + 9x$$

when  $t=0$ ,  $x=-2$  and  $v=-6$

$$(i) \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 18x^3 + 27x^2 + 9x$$

$$\frac{1}{2} v^2 = \frac{9}{2} x^4 + 9x^3 + \frac{9}{2} x^2 + C$$

When  $x=-2$ ,  $v=-6$  so

$$18 = 72 - 72 + 18 + C$$

$$C = 0$$

$$\therefore v^2 = 9x^4 + 18x^3 + 9x^2$$

$$v^2 = 9x^2(x^2 + 2x + 1)$$

$$v^2 = 9x^2(x+1)^2 \text{ as required.}$$

$$(ii) \text{ From } v^2 = 9x^2(x+1)^2$$

$$v = \pm 3x(x+1)$$

But when  $x=-2$ ,  $v=-6$  so

$$-6 = -3(-2)(-1)$$

$$\therefore v = -3x(x+1)$$

$$\therefore \frac{dx}{dt} = -3x(x+1)$$

$$\frac{dt}{dx} = \frac{1}{-3x(x+1)}$$

$$\int -3 dt = \int \frac{dx}{x(x+1)}$$

$$\therefore \int \frac{dx}{x(x+1)} = -3t$$

as required.

(iii)

$$\frac{dx}{dt} = -3x(x+1)$$

$$t = \int \frac{-1}{3x(x+1)} dx$$

$$t = -\frac{1}{3} \int \frac{1}{x} - \frac{1}{x+1} dx$$

$$t = -\frac{1}{3} [\ln x - \ln(x+1)] + C$$

But when  $t=0$ ,  $x=-2$  so

$$0 = -\frac{1}{3} \ln 2 + C$$

$$C = \frac{1}{3} \ln 2$$

$$\therefore t = -\frac{1}{3} \ln \left( \frac{x}{x+1} \right) + \frac{1}{3} \ln 2$$

$$t = \frac{1}{3} \ln \left( \frac{2(x+1)}{x} \right)$$

$$e^{3t} = \frac{2x+2}{x}$$

$$xe^{3t} = 2x+2$$

$$-2 = x(2 - e^{3t})$$

$$\therefore x = \frac{-2}{2 - e^{3t}}$$

$$(b) \int (1+2x^2) e^{x^2} dx$$

$$= \int e^{x^2} + 2x^2 e^{x^2} dx$$

$$= \int e^{x^2} dx + \int x(2xe^{x^2}) dx$$

$$= \int e^{x^2} dx + \left[ e^{x^2} \cdot x - \int e^{x^2} \cdot (1) dx \right]$$

$$= \int e^{x^2} dx + x e^{x^2} - \int e^{x^2} dx$$

$$= x e^{x^2}$$

(c) (i)

$$(\cos \theta + i \sin \theta)^7 = \cos 7\theta + i \sin 7\theta \text{ by de Moivre's Theorem}$$

$$\text{Also } (\cos \theta + i \sin \theta)^7 = \cos^7 \theta + 7 \cos^6 \theta (i \sin \theta) + 21 \cos^5 \theta (i \sin \theta)^2 + 35 \cos^4 \theta (i \sin \theta)^3 + 35 \cos^3 \theta (i \sin \theta)^4 + 21 \cos^2 \theta (i \sin \theta)^5 + 7 \cos \theta (i \sin \theta)^6 + (i \sin \theta)^7$$

by Binomial Theorem

$$= \{ \cos^7 \theta - 21 \cos^5 \theta \sin^2 \theta + 35 \cos^3 \theta \sin^4 \theta - 7 \cos \theta \sin^6 \theta \} + i \{ 7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta \}$$

$$\tan 7\theta = \frac{\sin 7\theta}{\cos 7\theta}$$

$$= \frac{7\cos^6\theta \sin\theta - 35\cos^4\theta \sin^3\theta + 21\cos^2\theta \sin^5\theta - \sin^7\theta}{\cos^7\theta - 21\cos^5\theta \sin^2\theta + 35\cos^3\theta \sin^4\theta - 7\cos\theta \sin^6\theta}$$

Divide numerator and denominator by  $\cos^7\theta$  to get

$$= \frac{7\tan\theta - 35\tan^3\theta + 21\tan^5\theta - \tan^7\theta}{1 - 21\tan^2\theta + 35\tan^4\theta - 7\tan^6\theta}$$

$$= \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6} \quad \text{if } t = \tan\theta.$$

$$(ii) \tan 7\theta = 0 \text{ if}$$

$$7t - 35t^3 + 21t^5 - t^7 = 0$$

$$-t(t^6 - 21t^4 + 35t^2 - 7) = 0$$

$$\therefore t = 0 \text{ or } t^6 - 21t^4 + 35t^2 - 7 = 0 \quad \text{--- (1)}$$

Let  $t^2 = x$ , then (1) becomes

$$x^3 - 21x^2 + 35x - 7 = 0 \quad \text{--- (2)}$$

The solutions to (2) lie among the solutions to

$$\tan 7\theta = 0$$

$$7\theta = 0, \pi, 2\pi, 3\pi, \dots, 6\pi$$

$$\theta = 0, \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7}, \dots, \frac{6\pi}{7}$$

$$\text{We let } x = t^2, \text{ so } x = \tan^2\theta$$

$\therefore$  the solutions to

$$x^3 - 21x^2 + 35x - 7 = 0 \text{ are}$$

$$x = \tan^2\left(\frac{\pi}{7}\right), \tan^2\left(\frac{2\pi}{7}\right), \text{ and } \tan^2\left(\frac{3\pi}{7}\right).$$

Q16, (a) (i)

To prove that  $\int_a^b g(x) dx = \int_a^b g(a+b-x) dx$ ,

consider  $I = \int_a^b g(a+b-x) dx$ .

$$\text{let } u = a+b-x$$

$$\therefore du = -dx$$

$$\text{when } x = b, \quad u = a$$

$$\text{when } x = a, \quad u = b$$

$$\therefore I = \int_b^a g(u) (-du)$$

$$= \int_a^b g(u) du$$

$$= \int_a^b g(x) dx$$

( $u$  is a dummy variable)

$$\therefore \int_a^b g(x) dx = \int_a^b g(a+b-x) dx$$

as required.

(b) (ii)

To show that  $\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$ .

where  $f(x)$  is an even function.

$$\text{Now } \int_{-a}^a \frac{f(x)}{1+e^x} dx$$

$$= \int_{-a}^a \frac{f(a+(-a)-x)}{1+e^{(a-a-x)}} dx$$

$$= \int_{-a}^a \frac{f(-x)}{1+e^{-x}} dx$$

$$= \int_{-a}^a \frac{e^x f(-x)}{1+e^x} dx$$

$$= \int_{-a}^a \frac{(1+e^x)f(-x)}{1+e^x} - \frac{f(-x)}{1+e^x} dx$$

$$= \int_{-a}^a f(-x) - \frac{f(-x)}{1+e^x} dx$$

$$= \int_{-a}^a f(x) - \frac{f(x)}{1+e^x} dx, \text{ since } f(x) = f(-x)$$

$$= \int_{-a}^a f(x) dx - \int_{-a}^a \frac{f(x)}{1+e^x} dx$$

$$\therefore 2 \int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_{-a}^a f(x) dx$$

$$\therefore 2 \int_{-a}^a \frac{f(x)}{1+e^x} dx = 2 \int_0^a f(x) dx$$

$$\therefore \int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx, \text{ as required.}$$

(b) Let  $I_n = \int_0^x t^n e^{-t} dt$ ,  $n \geq 0$  (integers)

(i) To prove that  $I_n = n! \left[ 1 - e^{-x} \left( 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) \right]$

when  $n=0$ ,  $I_n = \int_0^x t^0 e^{-t} dt$

$$= \int_0^x e^{-t} dt$$

$$= \left[ -e^{-t} \right]_0^x$$

$$= -e^{-x} - (-e^0) = 1 - e^{-x}$$

and also  $I_n = 0! \left[ 1 - e^{-x} (1) \right]$

$$= 1(1 - e^{-x}) = 1 - e^{-x}$$

Same!

$\therefore$  The statement is true when  $n=0$ .

Assume that the statement is true when  $n=k$ , i.e., assume

$$\int_0^x t^k e^{-t} dt = k! \left[ 1 - e^{-x} \left( 1 + x + \dots + \frac{x^k}{k!} \right) \right]$$

and then try to prove that

$$\int_0^x t^{k+1} e^{-t} dt = (k+1)! \left[ 1 - e^{-x} \left( 1 + x + \dots + \frac{x^k}{k!} + \frac{x^{k+1}}{(k+1)!} \right) \right]$$

Now  $\int_0^x t^{k+1} e^{-t} dt$

$$= \left[ -e^{-t} \cdot t^{k+1} \right]_0^x - \int_0^x -e^{-t} (k+1) t^k dt$$

$$= -x^{k+1} e^{-x} + (k+1) \int_0^x t^k e^{-t} dt$$

$$= -x^{k+1} e^{-x} + (k+1) \left[ k! \left( 1 - e^{-x} \left( 1 + x + \dots + \frac{x^k}{k!} \right) \right) \right]$$

by substituting the assumption.

$$= -x^{k+1} e^{-x} + (k+1)! \left[ 1 - e^{-x} \left( 1 + x + \dots + \frac{x^k}{k!} \right) \right]$$

$$= \frac{-x^{k+1} e^{-x} (k+1)!}{(k+1)!} + (k+1)! \left[ 1 - e^{-x} \left( 1 + x + \dots + \frac{x^k}{k!} \right) \right]$$

$$= (k+1)! \left[ 1 - e^{-x} \left( 1 + x + \dots + \frac{x^k}{k!} + \frac{x^{k+1}}{(k+1)!} \right) \right]$$

= RHS

$\therefore$  The statement is true by mathematical induction.

(ii) Show that

$$0 \leq \int_0^1 t^n e^{-t} dt \leq \frac{1}{n+1}$$

Now  $0 \leq e^{-t} \leq 1$  for  $0 \leq t \leq 1$

$$0 \leq t^n e^{-t} \leq t^n$$

$$0 \leq \int_0^1 t^n e^{-t} dt \leq \int_0^1 t^n dt$$

$$\text{But } \int_0^1 t^n dt = \left[ \frac{t^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1}$$

$$\therefore 0 \leq \int_0^1 t^n e^{-t} dt \leq \frac{1}{n+1} \text{ as required.}$$

(iii) sub  $x=1$  into the result in (i)  
and then use (ii) to get

$$0 \leq n! \left[ 1 - e^{-1} \left( 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} \right) \right] \leq \frac{1}{n+1}$$

$$0 \leq 1 - e^{-1} \left( 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} \right) \leq \frac{1}{(n+1)!}$$

(iv) As  $n \rightarrow \infty$ ,  
 $0 \leq \lim_{n \rightarrow \infty} \left[ 1 - e^{-1} \left( 1 + \dots + \frac{1}{n!} \right) \right] \leq \lim_{n \rightarrow \infty} \frac{1}{(n+1)!}$

$$\text{But } \lim_{n \rightarrow \infty} \frac{1}{(n+1)!} = 0$$

$\therefore$

$$\therefore \lim_{n \rightarrow \infty} [1 - e^{-1} (1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!})] = 0$$

$\therefore$  As  $n \rightarrow \infty$ ,

$$(1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}) \rightarrow e$$

because  $1 - e^{-1}(e) = 0.$